

Predicting Bank Loan Eligibility with Machine Learning : A Comparative Study of Advanced Algorithms

Sure Lakshmi Vyshnavi

Dept. of CSE

VFSTR Deemed to be University

Vadlamudi, Andhra Pradesh, India

lakshmivyshnavi4@gmail.com

D.S.V. Pavan

Dept. of CSE

VFSTR Deemed to be University

Vadlamudi, Andhra Pradesh, India

pavandasari744@gmail.com

T. Teja

Dept. of CSE

VFSTR Deemed to be University

Vadlamudi, Andhra Pradesh, India

tejachaitu264@gmail.com

Sadineni Sri Pujitha

Dept. of CSE

VFSTR Deemed to be University

Vadlamudi, Andhra Pradesh, India

sripujithasadineni@gmail.com

Syed Shareefunnisa

Dept. of CSE

VFSTR Deemed to be University

Vadlamudi, Andhra Pradesh, India

syedshareefa@gmail.com

Abstract—This paper presents a unique way for determining client loan eligibility utilizing powerful machine learning algorithms and data from several institutions. It focuses on the crucial phases of data purification, attribute selection, and comparative study of several machine learning approaches such as Gradient Boosting, CatBoost, XGBoost, AdaBoosting, and LightGBM. The study emphasizes the value of ensemble models in enhancing forecast accuracy and lowering default risk. Using cutting-edge machine learning techniques, this study offers banks and financial institutions with a powerful tool for improving loan eligibility checks. The suggested technique is shown to be reliable and effective through empirical validation and practical implementations.

Index Terms—Machine Learning - Gradient Boosting, Catboost, Xg boost, Adaboosting, lightGBM, Trained Dataset, Ensemble Models

I. INTRODUCTION

In the contemporary financial landscape, the allocation of loans represents a fundamental revenue stream for banks worldwide. However, the inherent risk associated with loan distribution necessitates a robust and reliable mechanism for assessing the creditworthiness of applicants. Traditional methods of loan evaluation often fall short in accurately predicting repayment capabilities, leading to increased default rates and financial losses for lending institutions. Recognizing this imperative, we introduce the Loan Prediction System Using Python, a novel application of machine learning algorithms to the domain of loan assessment. By harnessing the power of data analytics, this system aims to revolutionize the way banks evaluate loan applicants, thereby mitigating risk and optimizing lending practices. Through an exhaustive analysis of diverse variables including demographic information, financial history, and spending patterns, our model seeks to provide actionable insights into the creditworthiness of indi-

viduals. This research builds upon the rich legacy of regression analysis, tracing its origins back to the pioneering work of mathematicians such as Legendre, Gauss, and Markov in the early 19th century.

However, limitations of Linear Regression became apparent in scenarios where the relationship between variables was nonlinear or multidimensional. To address these challenges, researchers have developed advanced techniques such as polynomial regression and logistic regression, which offer greater flexibility in modeling complex relationships. Moreover, the emergence of big data technologies has enabled the analysis of large and diverse datasets, further enriching the capabilities of Machine Learning Algorithms. Here, we delve into methodology behind the Loan Prediction System, elucidating the process of data collection, model training, and evaluation.

II. RELATED WORK

Ravikumar et al. [12], delve into bankruptcy prediction in banks using ensemble classifiers.

Arun et al. [5], As the banking sector witnesses technological advancements and a surge in online loan applications, the authors highlight the importance of employing artificial intelligence algorithms to streamline the loan approval process.

A multi-round ensemble learning strategy based on heterogeneous ensemble frameworks is developed by Li et al. [10] to handle the challenge of precisely predicting default risk in peer-to-peer (P2P) lending. They emphasize how important it is to employ machine learning techniques in order to circumvent the limitations of traditional credit risk prediction models. The authors suggest a method that uses logistic regression (LR) and deep neural networks (DNN)

to augment extreme gradient boosting (XGBoost), the main modeling methodology, in order to further improve prediction accuracy.

By examining an actual bank credit dataset, Moro et al. [15] study the use of machine learning algorithms to forecast the success of bank telemarketing. Through their analysis, they identify 15 different machine learning algorithms that perform credibly well in predicting customers' ability to pay their loans. Additionally, they determine the most important features that influence creditworthiness and formulate a predictive model using linear regression based on these features.

Arun et al. [6] provide a technique for forecasting loan approval, highlighting the significance of machine learning in the banking sector. Through their analysis, they discover that several machine learning algorithms are suitable for forecasting a customer's loan eligibility, including decision trees, random forests, support vector machines, k-nearest neighbors, and decision trees with adaboosts.

The urgent problem of credit card default prediction in the face of unbalanced datasets is addressed by Alam et al. [3], which presents a big obstacle for commercial banks. Their results enhance the accuracy when utilizing their proposed methods, particularly with the Gradient Boosted Decision Tree (GBDT) method outperforming traditional machine learning classifiers.

A multi-stage self-adaptive classifier ensemble model is introduced by Guo et al. [4] with the goal of improving credit scoring prediction performance. Based on the features of the dataset, the model dynamically chooses basic classifiers from a Candidate Classifier Repository (CCR) and applies a Bayesian optimization technique to optimize the classifiers' parameters. In addition, the model uses the Particle Swarm Optimization (PSO) algorithm to calculate the final weights of basic classifiers and creates new features through multi-layer stacking.

To improve the accuracy of credit scoring categorization, Lawi et al. [8] suggest an Ensemble Logistic Regression model with Boosting approach utilizing the GradientBoost algorithm. Through their investigation, they show that the Ensemble GradientBoost method successfully raises the accuracy levels of a single Logistic Regression classifier on various datasets.

In their [1] work, Yang et al. present DeepCredit, a technique that uses user clickstream data to forecast loan risk in peer-to-peer (P2P) lending. With the proliferation of online lending platforms, assessing the creditworthiness of borrowers has become increasingly challenging. DeepCredit leverages deep learning techniques to analyze user clickstream data, providing insights into borrower behavior and enhancing loan risk prediction.

Aphale et al. [14] present a machine learning-driven loan approval system aimed at predicting loan application outcomes within the banking sector. By collecting user data including monthly income, marital status, loan amount, and duration, the system assists banks in determining loan approval decisions based on predefined parameters. The research highlights the significance of accurate forecasting in loan approval procedures, stressing crucial elements such as loan amount, length, age, income, zip code, and credit history in establishing the eligibility of applicants.

Li et al. [9] explore the application of the XGBoost (XGB) algorithm in personal credit evaluation, utilizing an open dataset from the Lending Club Platform. In their research, XGB is compared to different machine learning models, including Gradient Boosted Decision Trees (GBDT), Random Forests (RFs), Decision Trees (DT), and Logistic Regression (LR). The findings show that XGB outperforms the other models in terms of feature selection and classification accuracy. They highlight the potential for future research to extend this approach to multiclassification or regression problems within credit evaluation modeling.

A machine learning method for forecasting loan acceptance in the banking sector was presented by Aphale et al. [13], with a focus on cooperative banks. By utilizing machine learning algorithms, the suggested model is a viable way to improve loan approval procedures and reduce lending risks, especially in cooperative banking establishments.

Johnson et al. [11] delve into the pivotal issue of loan prediction, crucial for the financial sector's profitability. By analyzing statistics from Super Lender, an internet lending company, they validate the effectiveness of their predictive model. Their evaluation encompasses critical metrics such as F1-Score, Accuracy, Recall, and Precision, emphasizing the model's robustness.

A study by Smith et al. [2] aims to improve credit worthiness prediction models by applying preprocessing and feature selection steps. Evaluation criteria like F1-Score, Accuracy, and AUC show how effective their strategy is. The Random Forest (RF) and Decision Tree (DT) algorithms in particular show notable improvements in accuracy and F1-Score, with RF obtaining the greatest accuracy of 97.1.

Brown et al. [7] suggest a study that focuses on calculating loan settlement amounts based on geographic criteria in an attempt to improve accuracy within the banking system. In this, the new K-Nearest Neighbor (KNN) method is compared with the Support Vector Machine (SVM). Utilizing datasets from kaggle comprising 346 entries with 10 variables, the study employs advanced KNN algorithms to forecast, examine loan defaulters across multiple regions. The results show a significant difference ($p < 0.001$) between KNN and SVM, with KNN achieving an accuracy level of 71 and SVM only 52.

III. METHODOLOGY

A. Work Flow:

A machine learning pipeline is depicted in Fig. 1. and includes data gathering, preprocessing, visualization, splitting, and modeling with different boosting techniques. This figure explains model selection, evaluation techniques, data management, and visualization insights.

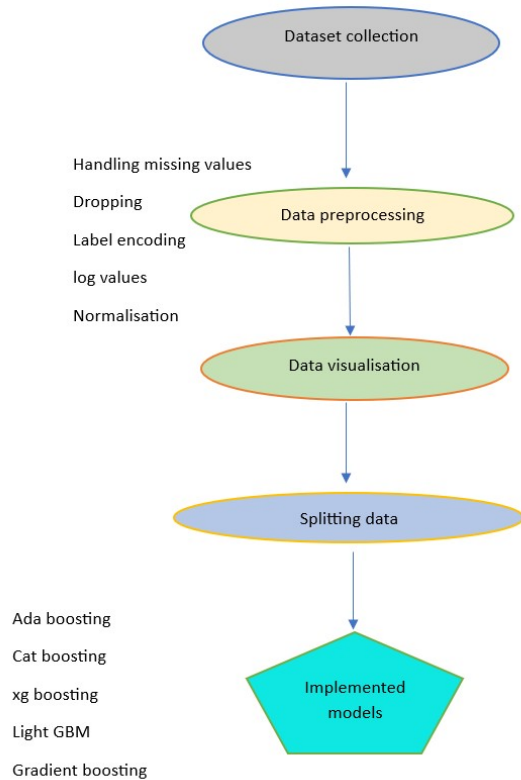


Fig. 1. Workflow

B. Dataset Collection:

Resource : The dataset is obtained from Kaggle and consists of 614 entries with 13 characteristics.

Attributes : Loan ID, Gender, Married, Dependents, Education, Self-Employed, Applicant Income, Coapplicant Income, Loan Amount, Loan Amount Term, Credit History, Property Location, Loan Status.

Nature : The dataset uses binary classification since the goal variable (Loan Status) has two unique values.

C. Preprocessing:

Handling Missing Values: For replacement, use the mean, median, or mode.

Dropping Columns: Remove any unneeded columns to ensure the right data structure.

Label Encoding: Convert category variables into unique, separate numeric values.

Log Transformation: Convert the Loan Amount to log numbers

for simplicity.

normalizing: Use the standard scaler for normalizing.

D. Data Visualizations:

Fig. 2. shows the dataset's visual representations, in the form of graphs and charts. These data visualizations provide important insights for additional study by highlighting important patterns, trends, or anomalies in the data.

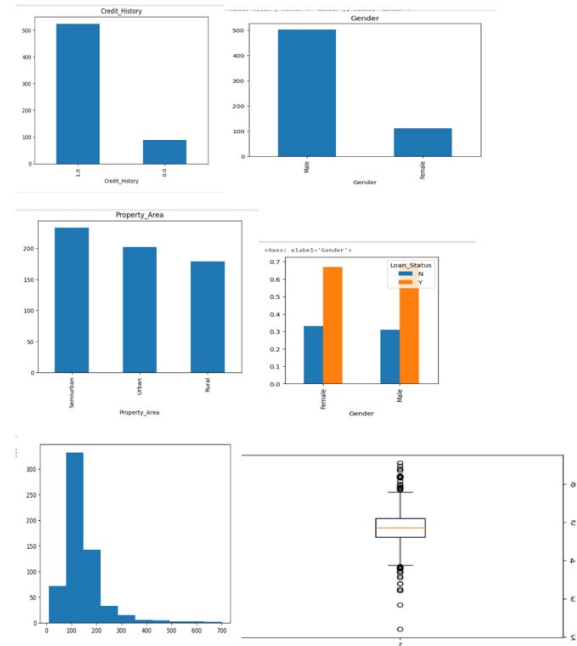


Fig. 2. Data Visualizations

E. Feature Extraction:

No feature extraction methods were utilized. Following preparation, the data was provided unchanged.

F. Splitting data:

Divided the dataset into 80% training and 20% testing.

G. Implemented Models:

1) Gradient boosting:

Iteratively constructs a collection of decision trees.

Each tree is taught to remedy the errors of prior trees.

Minimizes the loss function via gradient descent.

Hyperparameters:

n_estimators : 100

learning_rate : 0.1

max_depth : 3

min_samples : 2

min_samples_leaf : 1

subsample : 1.0

max_features : None

2)XGBoost:

It is an improved distributed gradient boosting library.

Handles both numerical and categorical data.

Sparse-aware split-finding algorithm.

Hyperparameters:

Number of estimators: 100
Learning rate: 0.1
Maximum depth: 3
Minimum child weight: 1
Gamma: zero
Subsample: 0.8
Column_by_tree: 0.8
Objective: 'binary'
Scale position weight: 1
Random state: 42

Xg Boosting Algorithm:

- 1: Initialize model prediction $F_0(x) = 0$ (baseline prediction)
- 2: **for** $m = 1$ to M (number of boosting stages) **do**
- 3: Compute negative gradients: $r_{im} = -\left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right]_{F(x)=F_{m-1}(x)}$ for all data points i
- 4: Construct a weak learner (e.g., shallow decision tree) to minimize the loss function on the negative gradients: $\hat{h}_m(x) = \arg \min_h \sum_{i=1}^N L(y_i, F_{m-1}(x_i) + h(x_i))$
- 5: Find the optimal step size: $\theta_m = \arg \min_{\theta} \sum_{i=1}^N L(y_i, F_{m-1}(x_i) + \theta \hat{h}_m(x_i))$
- 6: Update the model prediction: $F_m(x) = F_{m-1}(x) + \theta_m \hat{h}_m(x)$
- 7: **end for**
- 8: Output the final boosted model: $F_M(x)$

3)CatBoost:

Supports categorical data without preprocessing.
Implements ordered boosting and advanced gradient computing.
Excellent out-of-the-box performance.

Hyperparameters:

iterations: 1000
learning_rate: 0.1
depth: 6
l2_leaf_reg: 3
bootstrap_type: 'Bayesian'
loss_function: 'Logloss'
eval_metric: 'AUC'
cat_features: ['Gender', 'Married', 'Dependents', 'Education', 'Self-Employed', 'Property Area']
Random seed: 42

4)LightGBM:

Microsoft developed this software, which is recognized for its speed and efficiency.
A leaf-based tree development technique.
Handles big datasets.

Hyperparameters:

num_leaves: 31
max_depth: -1
learning_rate: 0.1
n_estimators: 100
objective: 'binary'

boosting_type: 'gbdt'
colsample_bytree: 0.8
subsample: 0.8
reg_alpha: 0.0
reg_lambda: 0.0
random_state: 42

5)AdaBoost:

combines numerous weak learners to create a powerful classifier.
Rebalances weights for misclassified occurrences in later rounds.

Hyperparameters:

n_estimators: 100
learning_rate: 1.0
base_estimator: DecisionTreeClassifier(max_depth=1)
algorithm: 'SAMME.R'
random_state: 42

Pseudo code:

Adaboost Algorithm:

- 1: Initialize weights $w_i = \frac{1}{m}$ for all data points (x_i, y_i)
- 2: **for** $t = 1$ to T **do**
- 3: Train a weak learner $h_t : X \rightarrow \{+1, -1\}$ based on the weighted distribution D_t
- 4: Compute the weighted error of h_t :
 $err_t = \sum_{i: h_t(x_i) \neq y_i} w_i$
- 5: **if** $err_t > \frac{1}{2}$ **then**
- 6: Terminate the loop (algorithm failed)
- 7: **end if**
- 8: $\alpha_t = \frac{1}{2} \ln \left(\frac{1 - err_t}{err_t} \right)$
- 9: Update weights for each data point:
 $w_i = w_i \exp(\alpha_t (-1)^{y_i h_t(x_i)})$
- 10: Normalize weights: $w_i = \frac{w_i}{\sum_j w_j}$
- 11: **end for**
- 12: Final hypothesis:
 $H(x) = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right)$

IV. PROPOSED MODEL

We employed ensembling models to predict bank loan eligibility. so, first we perform individually of 5 models. Those are catboosting, xgbboosting, LightGBM, Gradientboosting, Adaboosting. They give results as shown in TABLE(1). The highest accuracy among all these models is 78.86. But we decided to increase accuracy to some extent. So, we decided to fuse the models by all possible combinations. First we fuse two different models as shown in TABLE(2), TABLE(3) and then we fuse Three different combinations of models. They show results as TABLE(4). Then we fuse Four different combinations of models. Results in TABLE(5). To decide the highest accuracy among all these combinations we used voting classifier. It shows best accuracy among all possible combinations as 78.86%.

Initial Model Performance:

Individually tested models achieved accuracies of up to 78.86%.

models	accuracy	precision	recall	F1score
Gradientboosting	74.79	75.25	91.25	82.48
cataboosting	78.86	75.96	98.75	85.86
xgboosting	78.86	75.96	98.75	85.86
lightGBM	78.86	75.96	98.75	85.86
Adaboosting	78.86	75.96	98.75	85.86

TABLE I
COMPARISION BETWEEN 5 MODELS

c1	cataboosting
c2	xgbboosting
c3	LightGBM
c4	Gradientboosting
c5	Adaboosting

TABLE II
MODELS REPRESENTATION

Ensemble Learning:

Models can be combined in a variety of ways.

A voting classifier was used to achieve the highest level of accuracy.

K-Fold Validation:

Used 9-fold validation, resulting in 79.18% accuracy.

Applied 19-fold validation, achieving the best accuracy of 79.19%.

V. RESULTS :

A combination of XGBoost and AdaBoost with 19-fold validation provides the best accuracy when compared to others accuracy.

Evaluation Metrics:

- Average Accuracy: 0.7919799498746867
- Average Precision: 0.768796992481203
- Average Recall: 0.9868421052631579
- Average F1 Score: 0.8490728569675938

Two different models fusion	accuracy	precision	recall	F1score
c1c2	77.24	78.31	77.24	75.05
c1c3	78.05	80.72	78.05	75.42
c1c4	77.24	79.06	77.24	74.70
c1c5	78.05	80.72	78.05	75.42
c2c3	77.24	77.74	77.24	75.37
c2c4	74.80	74.50	74.80	72.90
c2c5	76.42	76.51	76.42	74.65
c3c4	76.42	76.95	76.42	74.33
c4c5	74.80	74.50	74.80	72.90

TABLE III
COMPARISION BETWEEN COMBINATIONS OF TWO DIFFERENT MODELS

Three different models fusion	accuracy	precision	recall	F1score
c1,c2,c3	78.86	80.48	78.86	76.83
c1,c2,c4	78.05	79.78	78.05	75.78
c1,c2,c5	77.24	78.31	77.24	75.42
c1,c3,c4	78.05	80.72	78.05	75.42
c1,c3,c5	78.86	82.53	78.86	76.15
c1,c4,c5	77.24	79.06	77.24	74.70
c2,c3,c4	76.42	76.95	76.42	74.33
c2,c3,c5	77.24	77.74	77.24	75.37
c3,c4,c5	78.86	81.37	78.86	76.51

TABLE IV
COMPARISION BETWEEN COMBINATIONS OF THREE DIFFERENT MODELS

Four different models fusion	accuracy	precision	recall	F1score
c1,c2,c3,c4	77.24	78.31	77.24	75.05
c1,c2,c3,c5	78.86	80.48	78.86	76.83
c1,c2,c4,c5	78.05	79.78	78.05	75.78
c1,c3,c4,c5	78.05	80.72	78.05	75.42
c2,c3,c4,c5	76.42	76.95	76.42	74.33

TABLE V
COMPARISION BETWEEN COIMBINATIONS OF FOUR DIFFERENT MODELS

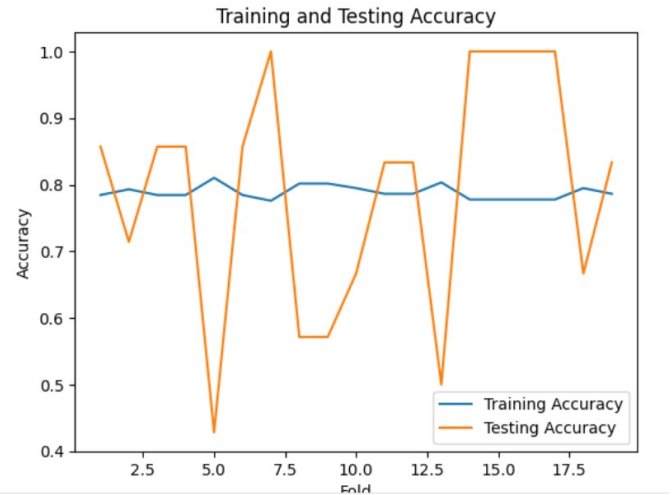


Fig. 3. Instruction and Assessment Precision curve of the optimal fusion model

The training and testing accuracy of the ideal fusion model throughout several epochs is shown in Fig. 3. Plotting during the training process shows variations in testing and training accuracy. Testing accuracy varies more than training accuracy, which normally rises. This raises the possibility of overfitting or underfitting problems that need more research.

Confusion matrix:

One instance of class 0 that was accurately predicted (true negatives) is displayed in the confusion matrix. An incorrect classification of a class 0 instance as a class 1 resulted in a false positive. True positives, or correct forecasts, were present in four of the class 1 cases. False negatives: Class 1 occurrences totaling 0 that were mispredicted as class 0. Given that it accurately forecasted every case in class 1 but misclassified one instance in class 0, the model appears to perform better for class 1 than class 0.

The model's capacity to accurately detect occurrences of class 0 is demonstrated by analyzing the true negatives and false positives. It also identifies locations where the model could mistake class 0 for class 1. Depending on the application's priorities, the high rate of true positives suggests great performance in reliably recognizing instances of class 1. False negatives, on the other hand, imply that there may be class 1 instances that the model is unable to identify. This might be a serious drawback in situations when it is necessary to correctly identify every instance of class 1.

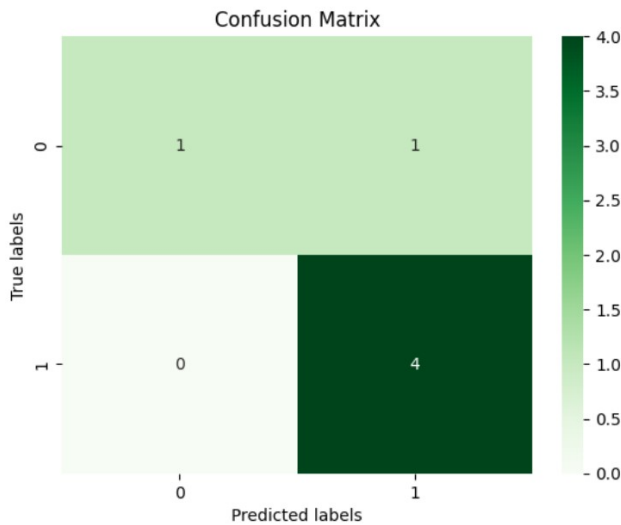


Fig. 4. Enter Caption

The model's confusion matrix, which illustrates the classification accuracy, is shown in Fig. 4. Four cases of the positive class (true positives) and one instance of the negative class (true negatives) were correctly identified by the model, according to the matrix. Nevertheless, it also incorrectly identified one positive class instance as false negative and one negative class instance as positive (false positive). These findings imply that the model's capacity to discriminate between the two classes with accuracy needs work.

Furthermore, a thorough grasp of the model's efficacy may be obtained by looking at the overall accuracy and precision metrics that are obtained from the confusion matrix.

VI. CONCLUSION:

The suggested technique, which includes preprocessing, model training, and ensemble learning, significantly increases the accuracy of bank loan eligibility forecasts. The use of modern machine learning algorithms and comprehensive validation assures strong and trustworthy findings, making it a helpful tool for financial organizations evaluating loan applications.

REFERENCES

- [1] Youness Abakarim, Mohamed Lahby, and Abdelbaki Attioui. Towards an efficient real-time approach to loan credit approval using deep learning. In *2018 9th International Symposium on Signal, Image, Video and Communications (ISIVC)*, pages 306–313. IEEE, 2018.
- [2] Akbar Affaruk Khuzaimi Ahmadani, Sinung Suakanto, Hanif Fakhrrur-roja, and Margareta Hardiyanti. Improving creditworthiness prediction using preprocessing stages and feature selection. In *2023 3rd International Conference on Electronic and Electrical Engineering and Intelligent System (ICE3IS)*, pages 195–200. IEEE, 2023.
- [3] Talha Mahboob Alam, Kamran Shaukat, Ibrahim A Hameed, Suhuai Luo, Muhammad Umer Sarwar, Shakir Shabbir, Jiaming Li, and Matloob Khushi. An investigation of credit card default prediction in the imbalanced datasets. *Ieee Access*, 8:201173–201198, 2020.
- [4] Shanshan Guo, Hongliang He, and Xiaoling Huang. A multi-stage self-adaptive classifier ensemble model with application in credit scoring. *IEEE Access*, 7:78549–78559, 2019.
- [5] Anshika Gupta, Vinay Pant, Sudhanshu Kumar, and Pravesh Kumar Bansal. Bank loan prediction system using machine learning. In *2020 9th International Conference System Modeling and Advancement in Research Trends (SMART)*, pages 423–426. IEEE, 2020.
- [6] Ch Naveen Kumar, D Keerthana, M Kavitha, and M Kalyani. Customer loan eligibility prediction using machine learning algorithms in banking sector. In *2022 7th international conference on communication and electronics systems (ICCES)*, pages 1007–1012. IEEE, 2022.
- [7] K Kalyan Kumar et al. Prediction of loan pricing on the basis of area location using k-nearest neighbour and support vector machine learning algorithms. In *2023 International Conference on Sustainable Communication Networks and Application (ICSCNA)*, pages 1036–1041. IEEE, 2023.
- [8] Armin Lawi, Firman Aziz, and Syafruddin Syarif. Ensemble gradient-boost for increasing classification accuracy of credit scoring. In *2017 4th international conference on computer applications and information processing technology (CAIPT)*, pages 1–4. IEEE, 2017.
- [9] Hua Li, Yumeng Cao, Siwen Li, Jianbin Zhao, and Yutong Sun. Xgboost model and its application to personal credit evaluation. *IEEE Intelligent Systems*, 35(3):52–61, 2020.
- [10] Wei Li, Shuai Ding, Yi Chen, and Shanlin Yang. Heterogeneous ensemble for default prediction of peer-to-peer lending in china. *Ieee Access*, 6:54396–54406, 2018.
- [11] P Nagaraj, K Nikhil, KVS Sai Ram Santosh Babu, D Hari Tejaswar Reddy, R Raja Sekar, and T Dhiliphan Rajkumar. Loan prediction analysis using innumerable machine learning algorithms. In *2023 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI)*, pages 1–6. IEEE, 2023.
- [12] Puvvala Ravikumar and Vadlamani Ravi. Bankruptcy prediction in banks by an ensemble classifier. In *2006 IEEE International Conference on Industrial Technology*, pages 2032–2036. IEEE, 2006.
- [13] Sk Sharmila, PVS Sandhya, P Suhani Kousar, P Anuradha, and S Deekshitha. Bank loan approval using machine learning. In *2024 International Conference on Integrated Circuits and Communication Systems (ICICACS)*, pages 1–5. IEEE, 2024.
- [14] Vishal Singh, Ayushman Yadav, Rajat Awasthi, and Guide N Partheeban. Prediction of modernized loan approval system based on machine learning approach. In *2021 international conference on intelligent technologies (CONIT)*, pages 1–4. IEEE, 2021.
- [15] Regina Esi Turkson, Edward Yeallakuor Baagyere, and Gideon Evans Wenyua. A machine learning approach for predicting bank credit worthiness. In *2016 Third International Conference on Artificial Intelligence and Pattern Recognition (AIPR)*, pages 1–7. IEEE, 2016.